

Notes: Kashyap, Rajan & Stein (JF, 2002)

Banks as Liquidity Providers: An Explanation for the Coexistence of Lending and Deposit-Taking

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
2.1	Contribution relative to theories of deposit-taking	3
2.2	Contribution relative to theories of lending	3
2.3	Contribution to liquidity-management theory	4
2.4	Contribution to empirical banking	4
2.5	Contribution to policy debates	4
3	Data and measurement	4
3.1	Historical motivation	4
3.2	NSSBF data for banks versus nonbanks	5
3.3	Call Report data for within-bank tests	5
3.4	Main variables	5
4	Models and theoretical specifications	5
4.1	Central economic idea	5
4.2	Model timeline	6
4.3	Liquid assets and costs	6
4.4	Deposit withdrawals and commitment takedowns	6
4.5	Optimal commitments and deposits in Region 1	6
4.6	Proposition 1	7
4.7	Proposition 2 and global comparative statics	7
4.8	Empirical predictions	7
5	Empirical specifications and identification design	7
5.1	Cross-institution comparison	7
5.2	Within-bank liquid-asset regression	8
5.3	Within-bank commitment regression	8
5.4	C&I commitment robustness test	8
5.5	Identification strengths	8
5.6	Identification limitations	9
6	Section-by-section study guide	9
6.1	Introduction	9
6.2	Historical overview	9

6.3	Model	9
6.4	Banks versus nonbank institutions	9
6.5	Within-bank evidence	9
6.6	Conclusion	9
7	Tables and figures	10
7.1	Figure 1: Liquid assets and commitments versus deposits	10
7.2	Table I: Banks versus Other Lending Institutions: Fraction of Each Institution's Lending Accounted for by Loan Type	10
7.3	Table II: Basic Descriptive Statistics for Deposits, Liquid Assets, and Commitments: Full Sample	12
7.4	Table III: Liquid Assets versus Transactions Deposits	12
7.5	Table IV: Commitments versus Transactions Deposits	13
7.6	Table V: Basic Descriptive Statistics for Deposits, Liquid Assets, and C&I Commitments: C&I Sample	13
7.7	Table VI: C&I Commitments versus Transactions Deposits	14
8	Result map	16
9	Short summary	16
10	Conclusion	16
10.1	Core conclusion	16
10.2	Economic intuition	17
10.3	Incremental contribution revisited	17
10.4	Implications for banking policy	17
10.5	Implications for monetary transmission	17
10.6	Limitations	17
10.7	Takeaway	17

1 Big picture

- **Question.** Why are deposit-taking and lending housed inside the same institution, rather than being split into a money-market mutual fund on one side and a finance company on the other?
- **Core answer.** The paper argues that deposits and loan commitments are two versions of the same primitive service: **liquidity on demand**. A depositor can withdraw funds unexpectedly; a borrower with a credit line can draw funds unexpectedly. Both create random liquidity outflows for the bank.
- **Main mechanism.** A bank that promises liquidity on demand must hold a costly buffer stock of liquid assets. If deposit withdrawals and loan-commitment takedowns are imperfectly correlated, one liquid-asset buffer can serve both sides of the balance sheet.
- **Synergy.** The bank is efficient because it shares a fixed liquidity-management infrastructure across two products. Deposits and commitments do not merely coexist historically; they can economize on the same costly liquidity stockpile.
- **Theory.** The model shows that, under relevant parameter regions, a higher demand-deposit franchise raises both liquid-asset holdings and the volume of loan commitments. The result requires that deposit and commitment outflows are not perfectly positively correlated.
- **Empirical predictions.** First, banks should do more commitment-based lending than nonbank lenders. Second, within banks, a higher transaction-deposit ratio should predict more liquid assets. Third, within banks, a higher transaction-deposit ratio should predict more loan commitments.
- **Evidence.** Using the 1993 National Survey of Small Business Finances and bank Call Report data from 1992–1996, the paper finds that banks are especially important providers of lines of credit, and that banks with more transaction deposits hold more liquid assets and issue more commitments.
- **Policy implication.** The paper challenges narrow-banking proposals that assume deposit-taking and lending are unrelated. If deposits and commitments share a liquidity-production technology, forcing them apart could destroy real efficiencies.
- **Economic takeaway.** Commercial banking is not just deposit-taking plus lending by accident. Its institutional form can be understood as a joint-production technology for supplying liquidity to households and firms.

2 Incremental contribution of the paper

2.1 Contribution relative to theories of deposit-taking

The deposit-taking literature, especially Diamond–Dybvig-style models and models of demandable debt, explains why demand deposits can be useful for households and investors. Those models often emphasize consumption insurance, monitoring incentives, bank runs, or liquidity creation. But by themselves, they do not explain why the same institution that issues demandable claims should also specialize in lending to firms.

Incremental contribution. Kashyap, Rajan, and Stein connect deposit-taking to a specific lending product: **loan commitments**. The paper shifts the question from “why are deposits useful?” to “why should deposits and commitments be jointly produced?” This is a stronger theory of why banks look different from money-market funds.

2.2 Contribution relative to theories of lending

The lending literature, especially delegated-monitoring models such as Diamond (1984), explains why loans to opaque borrowers may require intermediaries. But it does not necessarily explain why the intermediary must be funded with demand deposits rather than with bonds, equity, or wholesale debt.

Incremental contribution. The paper argues that what ties bank lending to bank liabilities is not ordinary term lending alone, but commitment-based lending. Once a bank grants a line of credit, the borrower effectively has a right to withdraw funds on demand. This makes a credit line economically similar to a demand deposit with a negative balance.

2.3 Contribution to liquidity-management theory

The paper contributes a joint-liquidity-management view of banks. The key scarce resource is not only borrower information or depositor trust, but a costly stock of liquid assets. A single liquidity buffer is more efficient when it covers multiple sources of outflow risk.

Incremental contribution. This is an across-balance-sheet diversification argument: banks diversify liquidity shocks between depositors and commitment borrowers. Prior work recognized diversification across depositors or borrowers; the novel point is diversification *across both sides of the balance sheet*.

2.4 Contribution to empirical banking

The empirical contribution is to connect the theory to measurable bank variables: transaction deposits, liquid assets, loan commitments, and C&I commitments. The paper uses both cross-institution evidence and within-bank cross-sectional evidence.

Incremental contribution. It develops direct tests of a liquidity-synergy model using observable balance-sheet variables. It shows that banks with a stronger transaction-deposit franchise are also more active in commitments, even after controlling for size and loan portfolio composition.

2.5 Contribution to policy debates

Narrow banking proposals often start from the premise that deposit-taking and lending are separable. The paper provides a reason this premise may be wrong.

Incremental contribution. By identifying a common liquidity-production function, the paper reframes narrow banking as a policy that may break up a real efficiency. The argument does not rely only on historical accident, deposit insurance distortions, or regulatory privileges.

3 Data and measurement

3.1 Historical motivation

- The paper begins with a historical overview of deposit banking and commitment-based lending.
- Deposit banking evolved from money changing, safekeeping, and payments services.
- Lending and credit lines evolved as ways to provide firms with flexible funding against uncertain liquidity needs.
- The history is not the formal test; it motivates the idea that banks have long combined payment/deposit services with liquidity provision to borrowers.

Interpretation: Historical continuity matters because it shows that the coexistence of deposits and lending is not just a modern artifact of regulation. The paper uses history to motivate a theory of functional complementarity.

3.2 NSSBF data for banks versus nonbanks

- To test whether banks provide more commitment-based lending than other institutions, the paper uses the 1993 National Survey of Small Business Finances.
- The sample covers more than 4,600 small businesses with fewer than 500 employees.
- Each loan is classified by loan type, such as unsecured line of credit, secured line of credit, mortgage, lease, auto loan, equipment loan, or other.
- Loans are also classified by the type of institution making the loan: commercial banks/S&Ls, finance/leasing companies, insurance companies, brokerage firms, mortgage banks, venture capitalists, and others.

Interpretation: This data set is used to test the cross-institution implication: if banks are liquidity providers, they should specialize in credit-line-like products relative to other lenders.

3.3 Call Report data for within-bank tests

- The main within-bank tests use quarterly Call Report data from 1992 through 1996.
- The unit is a bank holding company; banks within the same holding company are aggregated.
- The sample requires at least eight quarters of data.
- Banks are split into size categories: the 100 largest banks, the next 500 medium-sized banks, and remaining small banks.

Interpretation: Holding-company aggregation matters because liquidity management and commitment issuance may be coordinated within a bank holding company rather than only at individual-bank level.

3.4 Main variables

- **DEPRAT:** transaction deposits divided by total deposits. This proxies for the bank's deposit-taking franchise in demandable, payments-oriented claims.
- **LIQRAT:** cash plus securities divided by assets. This measures the bank's liquid-asset stockpile.
- **SECRAT:** securities divided by assets. This is a narrower liquid-asset measure that excludes cash.
- **COMRAT:** loan commitments divided by commitments plus loans. This measures how much of the bank's credit business is commitment-based.
- **CICOMRAT:** C&I commitments divided by C&I commitments plus C&I loans. This focuses on commercial and industrial credit commitments.

Interpretation: DEPRAT is the key explanatory variable because it captures the liability-side liquidity service. LIQRAT/SECRAT and COMRAT/CICOMRAT are the asset-side objects that should rise with deposits if the theory is correct.

4 Models and theoretical specifications

4.1 Central economic idea

The paper's model can be summarized by the following logic:

Demand deposits $\Rightarrow$ random withdrawals, Loan commitments $\Rightarrow$ random takedowns.

Both claims require the bank to hold liquid assets:

$$\text{Liquid-asset stockpile} = S_0.$$

Liquid assets are useful but costly. They may earn a low return, be subject to double taxation, or create agency problems because managers control financial slack.

Interpretation: The theory's central object is not the loan, the deposit, or the payment system separately. It is the costly capacity to meet unexpected customer liquidity demands.

4.2 Model timeline

- At date 0, the bank chooses term loans, liquid assets, demand deposits, and loan commitments.
- At date 1, deposit withdrawals and commitment takedowns may occur.
- If liquid assets are insufficient, the bank must raise costly external finance.
- At date 2, payoffs are realized.

Interpretation: The model uses a short horizon because the key trade-off is intertemporal liquidity management: holding liquid assets ex ante versus raising costly external funds ex post.

4.3 Liquid assets and costs

Let S_0 be liquid assets. Liquid assets earn the market rate i but impose a per-dollar deadweight cost τ . The net return between dates 0 and 1 is therefore:

$$(i - \tau)S_0.$$

Interpretation: The parameter τ is a reduced-form cost of liquidity. It can represent foregone interest on cash, tax costs of securities, agency costs of financial slack, or an equilibrium liquidity premium that makes liquid assets expensive to hold.

4.4 Deposit withdrawals and commitment takedowns

Let D_0 be demand deposits and C be commitments. At date 1, deposits may be withdrawn and commitments may be taken down. The paper allows these two outflow events to be correlated, but the key condition is:

$$\rho < 1.$$

That is, they are not perfectly positively correlated.

Interpretation: If the two outflow risks are imperfectly correlated, a single stock of liquid assets can serve both. This is the source of the joint-production synergy.

4.5 Optimal commitments and deposits in Region 1

In Region 1, liquid assets are scarce relative to both possible outflows, so:

$$S^* < \min(C^*, D_0).$$

The paper derives the first-order condition:

$$f + Cf_C = \frac{a[(2\rho - \rho^2 - 1)D_0 + (1 - \rho)C + 2(\tau/a)]}{4 - 2\rho}.$$

Implicit differentiation gives a positive comparative static:

$$\frac{dC^*}{dD_0} > 0 \quad \text{if} \quad \rho < 1.$$

Interpretation: In the central region, more deposits induce the bank to hold more liquidity. Because that extra liquidity can also support commitment takedowns, the bank optimally issues more commitments. This is the core theoretical result.

4.6 Proposition 1

Proposition 1. For parameters such that the equilibrium is in Region 1, both liquid-asset holdings S^* and commitments C^* are increasing in the amount of demand deposits D_0 .

Interpretation: The result does not say that deposits and commitments never compete. It says that when liquidity is scarce and outflow risks are not perfectly correlated, increasing deposits expands the bank's optimal liquidity buffer enough to make more commitments feasible.

4.7 Proposition 2 and global comparative statics

Proposition 2. As D_0 increases from zero to infinity, both S_0^* and C^* are weakly increasing and continuous functions. When liquidity costs are sufficiently large, equilibrium moves through regions where liquid assets rise and commitments either rise or plateau.

Interpretation: This proposition translates the local Region 1 intuition into a global cross-sectional prediction: banks with larger deposit franchises should be more likely to hold liquid assets and issue commitments.

4.8 Empirical predictions

The model yields three empirical predictions:

1. Deposit-taking banks offer relatively more commitments than other lending intermediaries.
2. Across banks, higher demand deposits predict higher liquid-asset holdings.
3. Across banks, higher demand deposits predict higher loan commitments.

Interpretation: These predictions move the paper from theory to observable balance-sheet patterns.

5 Empirical specifications and identification design

5.1 Cross-institution comparison

The first empirical test compares banks with nonbank lenders in the NSSBF:

$$\text{Loan-type share}_{j,k} = \frac{\text{Dollars of loan type } k \text{ from lender type } j}{\text{Total dollars of lending by lender type } j}.$$

Interpretation: If commercial banks are liquidity providers, they should have a disproportionately high share of lending in lines of credit, especially relative to institutions like mortgage banks, insurance companies, and venture capitalists.

5.2 Within-bank liquid-asset regression

The second test regresses liquid assets on transaction deposits:

$$LIQRAT_i = \alpha + \beta DEPRAT_i + \gamma' X_i + \varepsilon_i,$$

with a parallel specification for $SECRAT_i$.

Interpretation: A positive β supports the idea that demandable deposits require banks to hold larger liquid buffers. This is the empirical counterpart of Prediction 2.

5.3 Within-bank commitment regression

The third test regresses commitments on transaction deposits:

$$COMRAT_i = \alpha + \beta DEPRAT_i + \gamma' X_i + \varepsilon_i.$$

Controls include bank size, C&I loans relative to total loans, real estate loans relative to total loans, loans to individuals relative to total loans, and Federal Reserve district dummies.

Interpretation: A positive coefficient means that banks with more transaction deposits are also more commitment-oriented on the asset side. This is the empirical counterpart of Prediction 3.

5.4 C&I commitment robustness test

The paper also estimates:

$$CICOMRAT_i = \alpha + \beta DEPRAT_i + \gamma' X_i + \varepsilon_i,$$

where $CICOMRAT$ focuses on commercial and industrial commitments.

Interpretation: This addresses a key endogeneity concern. Total commitments may be related to total bank lending and balance-sheet composition. C&I commitments are narrower and less likely to be mechanically driven by overall loan demand.

5.5 Identification strengths

- The theory has multiple predictions across different data sets.
- The NSSBF test separates banks from nonbanks.
- The Call Report tests exploit cross-sectional variation within banks.
- Size controls and size-stratified estimates reduce the concern that results are only about large banks.
- The C&I commitment test addresses mechanical endogeneity between total loan demand and commitments.

5.6 Identification limitations

- DEPRAT is not randomly assigned; banks with transaction-deposit franchises may differ in unobserved ways.
- Commitment ratios and deposit ratios are both equilibrium choices.
- Call Report commitments measure unused commitments, not necessarily target commitments.
- The empirical tests are reduced-form; they do not estimate the structural parameters of the model.
- The model's liquidity synergy may coexist with other synergies, such as discount-window access, payments-system access, customer relationships, and information spillovers.

Interpretation: The evidence is not a randomized causal design. Its strength is that several observable patterns line up with the same liquidity-synergy mechanism.

6 Section-by-section study guide

6.1 Introduction

The introduction frames the puzzle: banks combine deposit-taking and lending. Existing theories explain either side separately, but not the joint institution. The authors propose that deposits and loan commitments are both liquidity-on-demand products.

6.2 Historical overview

The historical section motivates deposit-taking and lending as related bank functions. It is not formal evidence, but it shows that banks have historically combined payment services, deposit-taking, and credit provision.

6.3 Model

The model formalizes the synergy. The key mechanism is that liquid assets are costly, and imperfectly correlated deposit withdrawals and commitment takedowns allow one liquidity buffer to support both products.

6.4 Banks versus nonbank institutions

The NSSBF evidence shows that banks are the main providers of lines of credit. This supports the view that credit lines are a distinctively bank-like product.

6.5 Within-bank evidence

The Call Report evidence shows that banks with more transactions deposits hold more liquid assets and issue more commitments. The findings hold across size categories and in a C&I commitment robustness sample.

6.6 Conclusion

The conclusion emphasizes that banking may be efficient because lending and deposits share costly liquidity infrastructure. This has direct implications for narrow banking and the lending channel of monetary policy.

7 Tables and figures

7.1 Figure 1: Liquid assets and commitments versus deposits

Read:

- Panel A plots model-implied liquid assets against deposits.
- Liquid assets rise as deposits increase, reflecting the need to cover withdrawal risk.
- Panel B plots model-implied commitments against deposits.
- Commitments initially remain flat, rise over an intermediate range of deposits, and then flatten again.
- The rising region is the key joint-production region where extra deposit-related liquidity can also back commitment takedowns.

Detailed interpretation: Panel A is the direct Prediction 2 visual: a larger deposit franchise induces more liquidity holdings. Panel B is the direct Prediction 3 visual: a larger deposit franchise can induce more commitments. The figure is especially useful because it shows that the theory does not require commitments to rise with deposits everywhere. The relationship can be flat in regions where deposits do not relax the binding liquidity trade-off for commitments. The empirical prediction is therefore cross-sectional and broad rather than a knife-edge one-bank comparative static.

Economic message: The figure shows the core intuition of the paper: deposits and commitments can be complements because they share a costly liquidity buffer.

7.2 Table I: Banks versus Other Lending Institutions: Fraction of Each Institution's Lending Accounted for by Loan Type

Read:

- Commercial banks/S&Ls allocate 0.70 of their small-business lending dollars to total lines of credit.
- Finance/leasing companies also use lines of credit, but their line-of-credit share is lower at 0.51.
- Insurance companies, brokerage firms, mortgage banks, and venture capitalists have much lower line-of-credit shares.
- Mortgage banks specialize almost entirely in mortgages; venture capitalists are almost entirely in the "other" category.
- The table separates unsecured and secured lines of credit; commercial banks are important in both.

Detailed interpretation: This table is the paper's cross-institution evidence. It shows that banks are not merely generic lenders. They are especially active in the specific type of lending that resembles deposits: commitments and lines of credit. This is important because the theory is not about all lending; it is about commitment-based lending as a liquidity product. If banks only specialized in ordinary term loans, the paper's theory would be less compelling.

Economic message: Banks dominate the product category that requires on-demand liquidity provision. This is the first empirical sign that deposits and commitments belong together.

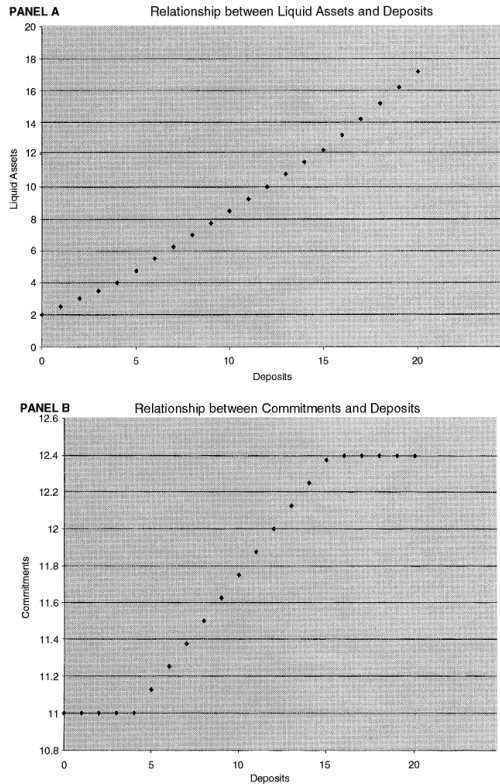


Figure 1. Liquid assets and commitments versus deposits. The figure plots the solution to the model for the following parameter values: $f(C) = 1 - 0.025C$; $\rho = 0.5$; $\alpha = 0.1$; and $\tau = 0.45$. The function $f(C)$ represents the fee on commitments; C is the volume of commitments issued; ρ is the correlation between commitment outflows and deposit outflows; α is a measure of the flow costs of external finance; and τ is a measure of the deadweight costs of holding financial slack. Panel A shows the relationship between liquid assets and deposits. Panel B shows the relationship between commitments and deposits.

Figure 1: Liquid assets and commitments versus deposits

Table I
Banks versus Other Lending Institutions: Fraction of Each Institution's Lending
Accounted for by Loan Type (by Dollar Volume)

The data in this table come from the 1993 National Survey of Small Business Finances, and cover a sample of 4,637 for-profit, nonfinancial, nonfarm enterprises that had fewer than 500 employees as of year-end 1992. Each loan is sorted into bins based on its type (line of credit, mortgage, etc.) and the type of financial institution it comes from. The loans in each bin are added up across all firms. The fraction of a particular loan type in an institution's portfolio is obtained by dividing the total loans in a bin by the total loans made by that type of institution.

	Unsecured Lines of Credit	Secured Lines of Credit	Total Lines of Credit	Leases	Mortgages	Auto Loans	Equipment Loans	Other
Commercial banks/S&Ls	0.31	0.39	0.70	0.02	0.13	0.01	0.06	0.07
Finance/leasing company	0.05	0.45	0.51	0.15	0.08	0.10	0.11	0.01
Insurance company	0.11	0.05	0.16	0.00	0.51	0.00	0.00	0.33
Brokerage firm	0.01	0.12	0.13	0.00	0.65	0.00	0.00	0.22
Mortgage bank	0.00	0.00	0.00	0.00	0.98	0.00	0.00	0.02
Venture capitalist	0.00	0.03	0.03	0.00	0.01	0.00	0.00	0.96
Other	0.01	0.02	0.04	0.04	0.34	0.01	0.09	0.49

Table I: Banks versus Other Lending Institutions

7.3 Table II: Basic Descriptive Statistics for Deposits, Liquid Assets, and Commitments: Full Sample

Read:

- The full Call Report sample contains 9,262 bank holding-company observations.
- Median LIQRAT is about 0.401 for all banks, meaning cash plus securities are a substantial part of bank assets.
- Median DEPRAT is about 0.281 for all banks, showing meaningful variation in transactions deposits.
- Median COMRAT is about 0.078 for all banks, with higher values in some size categories.
- The table separates large, medium-sized, and small banks to show that scale affects the distribution of deposits, liquidity, and commitments.

Detailed interpretation: Table II is not a causal test. It establishes that the variables used in the paper have substantial cross-sectional variation. This matters because the theory predicts relationships among variables: transaction deposits, liquid assets, and commitments. Without meaningful variation in DEPRAT or COMRAT, the empirical tests would be weak. The table also reveals why the paper controls for size: large banks, medium banks, and small banks differ in balance-sheet composition.

Economic message: The descriptive statistics show that banks differ substantially in both their liability-side demandability and asset-side liquidity/commitment profiles, creating the cross-sectional variation needed to test the theory.

Table II
Basic Descriptive Statistics for Deposits, Liquid Assets, and Commitments: Full Sample

The sample begins with all federally insured banks between 1992 and 1996. All banks within the same holding company are aggregated together to form a single holding-company-level observation, and observations for which there are less than eight quarters of data are then dropped. Size categories are based on the average level of real total assets: Large banks are the 100 banks with the highest values of average real assets; medium-sized banks are the next 500 banks; and small banks are the remaining banks. LIQRAT is the ratio of cash plus securities to assets, SECRAT is the ratio of securities to assets, DEPRAT is the ratio of transactions deposits to total deposits, COMRAT is the ratio of loan commitments to loan commitments plus loans. Q1 indicates the first quartile, Med. refers to the median, and Q3 denotes the third quartile.

1992-1996 Time-averaged Data	All Banks (N = 9,262)			Large Banks (N = 100)			Medium-sized Banks (N = 500)			Small Banks (N = 8,662)		
	Q1	Med.	Q3	Q1	Med.	Q3	Q1	Med.	Q3	Q1	Med.	Q3
LIQRAT	0.319	0.401	0.511	0.264	0.313	0.369	0.293	0.369	0.453	0.322	0.404	0.515
SECRAT	0.262	0.344	0.454	0.198	0.249	0.302	0.244	0.318	0.399	0.264	0.347	0.458
DEPRAT	0.223	0.281	0.345	0.241	0.293	0.327	0.167	0.263	0.327	0.225	0.282	0.346
COMRAT	0.041	0.078	0.127	0.186	0.260	0.333	0.086	0.134	0.180	0.039	0.075	0.120

Table II: Descriptive statistics, full sample

Table III
Liquid Assets versus Transactions Deposits

Dependent variable in Panel A is LIQRAT, the ratio of cash plus securities to assets. In Panel B, the dependent variable is SECRAT, the ratio of securities to assets. Each cell displays the point estimate, *t*-statistic, and "explanatory power" for the OLS coefficient on DEPRAT, the ratio of transactions deposits to total deposits. Explanatory power is defined as the coefficient on DEPRAT, times the standard deviation of DEPRAT, divided by the standard deviation of the dependent variable. The other independent variables whose coefficients are not reported include the log of real bank assets, the ratio of commercial and industrial loans to total loans, the ratio of real estate loans to total loans, the ratio of loans to individuals to total loans, and Federal Reserve district dummies. The sample begins with all federally insured banks between 1992 and 1996. All banks within the same holding company are aggregated together to form a single holding-company-level observation, and observations for which there are less than eight quarters of data are then dropped. Size categories are based on the average level of real total assets: Large banks are the 100 banks with the highest values of average real assets; medium-sized banks are the next 500 banks; and small banks are the remaining banks.

	Sample			
	All Banks	Large Banks	Medium-sized Banks	Small Banks
Number of observations	9,262	100	500	8,662
Panel A: Dependent Variable = LIQRAT: (cash + securities)/assets				
Coefficient on DEPRAT	0.227	0.313	0.105	0.235
(<i>t</i> -statistic)	(16.62)	(2.94)	(1.82)	(16.67)
Explanatory power of DEPRAT	0.172	0.309	0.084	0.177
Panel B: Dependent Variable = SECRAT: securities/assets				
Coefficient on DEPRAT	0.153	0.255	0.056	0.158
(<i>t</i> -statistic)	(10.83)	(2.64)	(0.95)	(10.84)
Explanatory power of DEPRAT	0.114	0.280	0.045	0.118

Table III: Liquid Assets versus Transactions Deposits

7.4 Table III: Liquid Assets versus Transactions Deposits

Read:

- The dependent variables are LIQRAT in Panel A and SECRAT in Panel B.
- For all banks, the coefficient of DEPRAT on LIQRAT is 0.227 with a *t*-statistic of 16.62.

- For all banks, the coefficient of DEPRAT on SECRAT is 0.153 with a t -statistic of 10.83.
- The explanatory power of DEPRAT is economically meaningful: 0.172 for LIQRAT and 0.114 for SECRAT in the all-bank sample.
- The positive relationship is especially strong for small banks and remains positive for large and medium-sized banks, though precision varies with sample size.

Detailed interpretation: Table III directly tests Prediction 2. The coefficient on DEPRAT asks whether banks with more transactions deposits hold more liquid assets after controlling for bank size, loan mix, and district dummies. The answer is yes. The SECRAT specification is important because it shows that the result is not only about cash; securities also rise with transactions deposits. In the theory, this liquid-asset buffer is the common resource that supports both deposit withdrawals and commitment takedowns.

Economic message: Banks with a stronger demand-deposit franchise hold larger liquid-asset buffers. This supports the model's premise that deposits require costly liquidity stockpiles.

7.5 Table IV: Commitments versus Transactions Deposits

Read:

- The dependent variable is COMRAT, the ratio of loan commitments to commitments plus loans.
- For all banks, the coefficient on DEPRAT is 0.116 with a t -statistic of 20.21.
- The coefficient is positive across large, medium-sized, and small banks.
- Explanatory power is 0.181 for all banks, 0.207 for large banks, 0.242 for medium-sized banks, and 0.188 for small banks.
- The relationship remains strong despite controls for size, loan composition, and Federal Reserve district.

Detailed interpretation: Table IV tests Prediction 3. It asks whether the same banks that specialize in transactions deposits also specialize in loan commitments. The positive coefficient supports the paper's central claim that the two products are complementary. The strongest interpretive point is not simply that bigger banks do both; the regression controls for size, and the relationship persists within size classes.

Economic message: Deposit-taking and commitment-based lending appear linked inside the banking sector. This is the main empirical evidence for joint liquidity production.

7.6 Table V: Basic Descriptive Statistics for Deposits, Liquid Assets, and C&I Commitments: C&I Sample

Read:

- The C&I sample contains 4,257 all-bank observations, smaller than the full sample because it restricts banks by C&I loan shares.
- CICOMRAT replaces COMRAT and focuses on commercial and industrial commitments relative to C&I lending.
- Median CICOMRAT is about 0.251 for all banks and is higher among large banks.
- LIQRAT, SECRAT, and DEPRAT continue to show substantial variation in the restricted sample.

Table IV

Commitments versus Transactions Deposits

Dependent variable is COMRAT, the ratio of loan commitments to loan commitments plus loans. Each cell displays the point estimate, t -statistic, and "explanatory power" for the OLS coefficient on DEPRAT, the ratio of transactions deposits to total deposits. Explanatory power is defined as the coefficient on DEPRAT, times the standard deviation of DEPRAT, divided by the standard deviation of the dependent variable. The other independent variables whose coefficients are not reported include the log of real bank assets, the ratio of commercial and industrial loans to total loans, the ratio of real estate loans to total loans, the ratio of loans to individuals to total loans, and Federal Reserve district dummies. The sample begins with all federally insured banks between 1992 and 1996. All banks within the same holding company are aggregated together to form a single holding-company-level observation, and observations for which there are less than eight quarters of data are then dropped. Size categories are based on the average level of real total assets: Large banks are the 100 banks with the highest values of average real assets; medium-sized banks are the next 500 banks; and small banks are the remaining banks.

	Sample			
	All Banks	Large Banks	Medium-sized Banks	Small Banks
Number of observations	9,262	100	500	8,662
Coefficient on DEPRAT	0.116	0.232	0.160	0.113
(t -statistic)	(20.21)	(3.35)	(5.80)	(19.35)
Explanatory power of DEPRAT	0.181	0.207	0.242	0.188

Table IV: Commitments versus Transactions Deposits

Table V

Basic Descriptive Statistics for Deposits, Liquid Assets, and C&I Commitments: C&I Sample

The sample begins with all federally insured banks between 1992 and 1996. All banks within the same holding company are aggregated together to form a single holding-company-level observation, and observations for which there are less than eight quarters of data are then dropped. Size categories are based on the average level of real total assets: Large banks are the 100 banks with the highest values of average real assets; medium-sized banks are the next 500 banks; and small banks are the remaining banks. The sample is then further screened to remove any banks whose ratio of C&I loans to total loans is less than 10 percent or greater than 50 percent. LIQRAT is the ratio of cash plus securities to assets. SECURAT is the ratio of securities to assets. DEPRAT is the ratio of transactions deposits to total deposits. CICOMRAT is the ratio of C&I commitments to C&I commitments plus C&I loans. Q1 indicates the first quartile, Med. refers to the median, and Q3 denotes the third quartile.

1992-1996 Time-averaged Data	All Banks ($N = 4,257$)			Large Banks ($N = 80$)			Medium-sized Banks ($N = 265$)			Small Banks ($N = 3,912$)		
	Q1	Med.	Q3	Q1	Med.	Q3	Q1	Med.	Q3	Q1	Med.	Q3
LIQRAT	0.322	0.397	0.498	0.273	0.313	0.376	0.310	0.387	0.452	0.325	0.399	0.503
SECURAT	0.263	0.338	0.441	0.208	0.251	0.301	0.258	0.330	0.394	0.265	0.341	0.446
DEPRAT	0.248	0.301	0.361	0.260	0.302	0.335	0.252	0.300	0.358	0.247	0.301	0.362
CICOMRAT	0.149	0.251	0.350	0.458	0.526	0.575	0.270	0.364	0.439	0.140	0.241	0.334

Table V: Descriptive statistics, C&I sample

Detailed interpretation: Table V sets up the robustness test in Table VI. The C&I sample is important because total commitments can include categories that do not map neatly to the theoretical notion of business credit lines. By restricting to banks with meaningful C&I lending and using CICOMRAT, the paper narrows the outcome to the type of commitment most closely related to corporate liquidity demand.

Economic message: The restricted C&I sample shows that the paper's commitment result is not driven only by broad or poorly measured commitment categories.

7.7 Table VI: C&I Commitments versus Transactions Deposits

Read:

- The dependent variable is CICOMRAT.
- For all banks, the coefficient on DEPRAT is 0.209 with a t -statistic of 9.64.
- The coefficient is positive for large banks, medium-sized banks, and small banks.
- The point estimates for large and medium-sized banks are larger than the all-bank coefficient, though sample sizes are smaller.
- The explanatory power of DEPRAT remains economically meaningful, especially for large and medium-sized banks.

Detailed interpretation: Table VI is the paper's strongest robustness check against an endogeneity story in which general loan demand mechanically drives both deposits and commitments. Because C&I commitments are a narrower category and the sample screens out banks with extreme C&I loan shares, the positive DEPRAT coefficient is harder to dismiss as a broad balance-sheet artifact. The result shows that transaction deposits are linked to business credit commitments specifically, which is exactly what the liquidity-synergy model predicts.

Economic message: The C&I evidence reinforces the paper's main message: banks with stronger transaction-deposit franchises are better positioned to provide business liquidity commitments.

Table VI
C&I Commitments versus Transactions Deposits

Dependent variable is CICOMRAT, the ratio of C&I commitments to C&I commitments plus C&I loans. Each cell displays the point estimate, *t*-statistic, and "explanatory power" for the OLS coefficient on DEPRAT, the ratio of transactions deposits to total deposits. Explanatory power is defined as the coefficient on DEPRAT, times the standard deviation of DEPRAT, divided by the standard deviation of the dependent variable. The other independent variables whose coefficients are not reported include the log of real bank assets, the ratio of commercial and industrial loans to total loans, the ratio of real estate loans to total loans, the ratio of loans to individuals to total loans, and Federal Reserve district dummies. The sample begins with all federally insured banks between 1992 and 1996. All banks within the same holding company are aggregated together to form a single holding-company-level observation, and observations for which there are less than eight quarters of data are then dropped. Size categories are based on the average level of real total assets: Large banks are the 100 banks with the highest values of average real assets; medium-sized banks are the next 500 banks; and small banks are the remaining banks. The sample is then further screened to remove any banks whose ratio of C&I loans to total loans is less than 10 percent or greater than 50 percent.

	Sample			
	All Banks	Large Banks	Medium-sized Banks	Small Banks
Number of observations	4,257	80	265	3,912
Coefficient on DEPRAT	0.209	0.389	0.406	0.198
(<i>t</i> -statistic)	(9.64)	(2.62)	(4.39)	(8.80)
Explanatory power of DEPRAT	0.127	0.283	0.246	0.128

Table VI: C&I Commitments versus Transactions Deposits

8 Result map

- **Historical motivation:** banks historically combine payments, deposits, and credit because these functions share liquidity and information infrastructure.
- **Model result 1:** in the key region, higher demand deposits increase both liquid assets and commitments.
- **Model result 2:** globally, liquid assets and commitments are weakly increasing in deposits under the model's conditions.
- **Prediction 1 evidence:** commercial banks/S&Ls have the highest share of line-of-credit lending in the NSSBF data.
- **Prediction 2 evidence:** DEPRAT positively predicts LIQRAT and SECRAT in Call Report data.
- **Prediction 3 evidence:** DEPRAT positively predicts COMRAT in the full bank sample.
- **C&I robustness:** DEPRAT positively predicts CICOMRAT in the restricted C&I sample.
- **Policy implication:** the bank balance sheet has an economic logic that narrow banking may miss.

9 Short summary

- The paper asks why banks combine deposit-taking and lending.
- The authors argue that demand deposits and loan commitments are both claims on liquidity on demand.
- Both products require the bank to hold a costly liquid-asset stockpile.
- If deposit withdrawals and commitment takedowns are imperfectly correlated, one liquidity buffer can support both products.
- The model predicts that more demand deposits should increase both liquid assets and commitments.
- The paper tests the theory using the NSSBF and Call Report data.
- The NSSBF shows that banks specialize in lines of credit relative to other lenders.
- Call Report regressions show that transaction deposits predict liquid-asset holdings.
- Call Report regressions also show that transaction deposits predict loan commitments.
- The C&I commitment analysis confirms the result in a narrower business-lending sample.
- The paper's contribution is to show that the coexistence of lending and deposit-taking may be efficient, not accidental.
- The policy message is that narrow banking proposals should consider the joint-production benefits of banks' liquidity services.

10 Conclusion

10.1 Core conclusion

The paper's core conclusion is that banks exist as joint producers of liquidity. Deposit-taking and commitment-based lending are not two unrelated balance-sheet activities. They are linked because both allow customers to demand funds unpredictably, and both require the bank to hold costly liquid assets.

10.2 Economic intuition

The easiest way to understand the paper is to think of a credit line as a reverse deposit. A depositor has the right to withdraw positive funds; a borrower with a commitment has the right to withdraw credit. Both rights create liquidity risk for the bank. If these risks are not perfectly correlated, one liquid-asset buffer can serve both. This generates economies of scope.

10.3 Incremental contribution revisited

The paper’s incremental contribution is to build a theory of the bank as a liquidity platform. Prior theories explain deposits or loans separately. Kashyap, Rajan, and Stein explain why commitments and deposits belong together. The empirical contribution is to show that the predicted link appears in both cross-institution and within-bank data.

10.4 Implications for banking policy

The results matter for debates about narrow banking. If deposits and lending are unrelated, it may be efficient to separate them. But if deposits and commitments share liquidity infrastructure, separation may destroy efficiencies. This does not mean all forms of bank regulation are inefficient; it means that policy should not ignore the economic role of balance-sheet integration.

10.5 Implications for monetary transmission

The paper also matters for the bank lending channel. If banks provide commitments because deposits support liquidity management, then changes in reserves or deposit funding can affect the supply of credit commitments. This connects the liability side of banks to corporate liquidity provision.

10.6 Limitations

The empirical tests are reduced-form and cross-sectional. DEPRAT is not randomly assigned, and the model’s structural parameters are not estimated. The paper also focuses on explicit loan commitments, while many bank relationships may involve implicit commitments. Finally, bank synergies may also involve payments-system access, discount-window access, information reuse, and relationship spillovers.

10.7 Takeaway

The enduring lesson is that banks are not just lenders that happen to take deposits. They are institutions specialized in supplying liquidity on demand to multiple customer groups. The coexistence of deposits and commitments is therefore a central feature of banking, not an accident.